VU VRIJE UNIVERSITEIT AMSTERDAM

De-identification of Qualitative and Quantitative data

Research Policy Support (RePS) - FGB

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Key concepts

Key concepts: Direct and Indirect identifiers

Direct identifiers

Information that can be directly traced back to the participant

Examples:

- Names
- Home address
- Personal E-mail
- Video or photographs of faces

Strong indirect identifiers

The distinction between direct and indirect identifiers is not always very clear, hence this category.

Examples:

- Phone number
- Date of birth
- BSN !!

Indirect identifiers

Characteristics that, when cross-referenced with other data, can result in the re-identification of an individual.

Examples:

- Age
- Gender
- Occupational information / workplace
- Education history
- Income
- Geographical location
- IP address
- Exact dates

Key concepts: Anonymisation VS Pseudonymisation

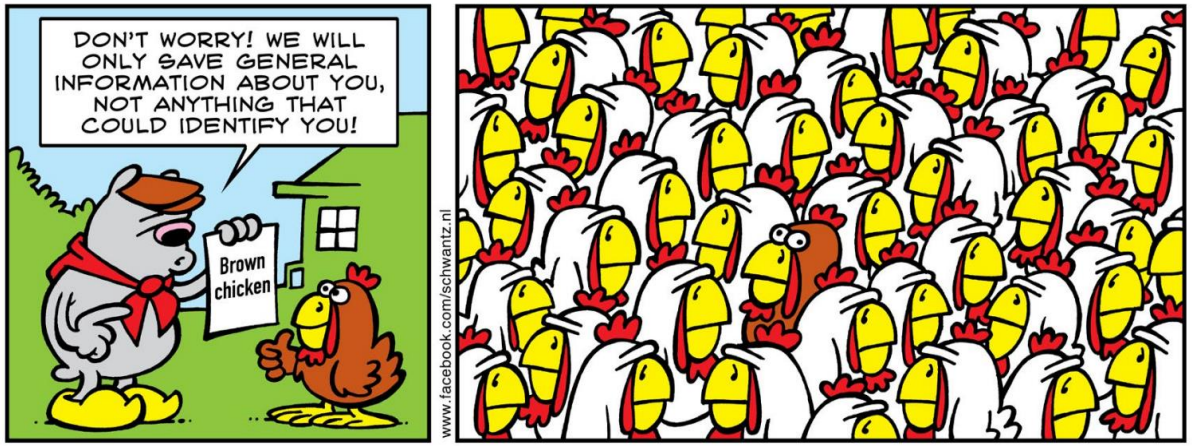
When in doubt, contact RePS to determine whether your data can be considered fully anonymous

Anonymised data

- It is not possible to identify an individual through direct or indirect identifiers
- Anonymized data is **no longer subject to the GDPR**

Pseudonymous data

- The data can be linked back to the individual if you have additional information (a key or a separate database)
- Pseudonymous data is still personal data, hence **subject to the GDPR**



Key concepts: Anonymisation VS Pseudonymisation

PERSONAL DATA		ANONYMOUS DATA
Fully identifiable data	*Pseudonymous data*	
Any data from which a person could immediately be identified with little to no effort (e.g. names, faces, unique characteristics).	Data from which a person cannot be immediately identified, however it remains possible to identify a person from the data.	Data which cannot be traced back to an individual person. Data collected from human subjects, either directly or when re-used for secondary research, is ***rarely ever*** anonymous.
EXAMPLE		
Patient number 90210 City Leeuwarden Date of birth 27-4-1967 Income 7.861 Job Judge Car DeLorean License Plate SN-09-HN	Study subject 47110009 Region Friesland Year of birth 1967 Income 7.500-10.000 Job Legal Car DeLorean	Country Netherlands Age 51-60 Income 5.000 – 15.000 Job Legal Car sportwagen

WANT TO KNOW MORE?
TURN CARD OVER!

National Coordination Point Research Data Management

DOI 10.5281/zenodo.3584842

Example

Doe v. Netflix

- In 2006 Netflix released an allegedly anonymous database containing 10 million movie rankings made by half a million users.
- Data was released for a contest of Netflix, where participants attempted to improve the recommendation algorithm of Netflix
- Database could be linked with another open online movie review database (IMDb) by comparing the ratings with public scores on the IMDb film website in 2014
- Revelations included identifying their political leanings and sexual orientation.
- In-the-closet lesbian mother sued Netflix for privacy invasion, alleging that the database made it possible for her to be outed.

De-identification techniques

1) Removing identifiers

Remove direct or indirect identifiers to decrease the risk of reidentification

Examples:

- Name
- Email
- Facial recognition
- Location identifiers
- Medical record numbers

2) Masking

- Suppression of strong identifiers
- Replacing certain numbers or letters with a * or **X**
- However, remaining identifying data could be recombined to re-identify persons.
  - E.g. the triple date of birth, gender and zip code is enough to uniquely identify at least 87% of US citizens in publicly available datasets

Phone number	Data of birth	Gender	ZIP code
06********	23/04/1987	Male	5673RF
06********	4/5/1999	Female	6203HG
06********	15/10/1995	Male	5623YF

3) Aggregating or recoding

Reduce the precision of variables, reducing into fewer categories

Original Data	Aggregated/recoded data
School name: Amsterdam International School	School type: Secondary School
Age: 15	Age range: 10-18
City/Region: Amsterdam	Country: Netherlands

- Occupation > [a starting position]
- Workplace > [a construction company]
- Eye surgery > [medical procedure]
- Names of persons > [sister], [mother], [colleague, female], etc
- AIDS > [long-term illness]

Removing hidden metadata from files

In case you are using data that was collected with the participants **personal devices**, it is important to check if there is any hidden **technical metadata** saved when the files were created that could enable de reidentification of participants.

- Location, information about the owner of a device or user profile

How to [remove metadata from Microsoft word]

e.g1. Textual data created and submitted directly by the participant containing metadata that refer to the participant explicitly

e.g2. A picture submitted with the exact coordinates of where they were taken and the photographer's name

How to de-identify your data?

Steps for deidentification

1.Identify the direct identifiers:

Which are part of the research analysis (e.g. facial expressions of participants) and which ones aren't (e.g. contact details of participants)

2. Assess if steps can be taken to reduce the identifiability of data (during collection).

E.g. if collecting audiovisual, is it possible to collect only audio and no video?

- Slightly less identifiable data (but still identifiable)

3. Separate the direct identifiers from the research data:

In case you could not separate the direct identifiers from the rest of the data during collection, now it is the time to do so.

4. Review and modify direct identifiers:

- Highly unique information
- When combining several variables together enables to reidentify an individual.

Is my data anonymous?

There are three guiding principles for anonymous data:

- It must **not** be possible to couple together a person's record across different datasets
- It must **not** be possible to assume, with a strong likelihood, specific information about a person based on the other information present in the dataset
- It must **not** be possible to single out at least one unique person in the dataset

> Check with your privacy champion and evaluate whether **the three conditions apply** before assuming that your data are now anonymized.

> The principles are not perfect, they require **contextual awareness**

Contextual awareness

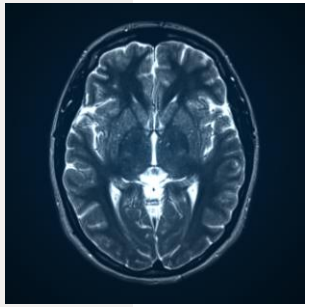
Evaluating the extent to which deidentification is needed requires **contextual awareness.**

1. Sensitivity of the data
2. Vulnerability of the participants
3. Outliers

E.g. A series of blood pressure measurements is very variable on a constant basis, which makes it much harder variables to serve as identifiers

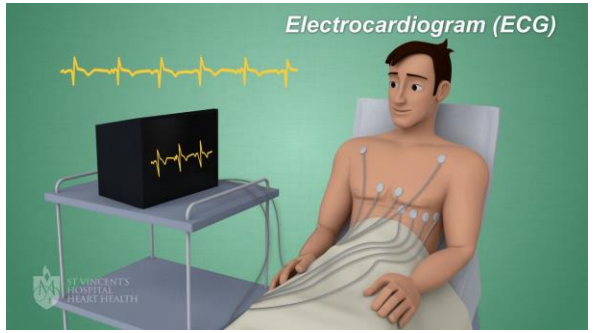
E.g.2. Eye tracking data can be used to reidentify the individual. However, that would require a significant effort, and the data available is not too sensitive.

How far can you de-identify the data before it becomes unusable?



When you start collecting any data related to the **body or health** of individuals, the data of these measurements becomes very difficult to de-identify because otherwise you would make the data unusable.

Take additional measures



Additional measures

Purpose limitation

Multiple de-identification measures

Appropriate storage location

De-identification techniques do not exist in a bubble

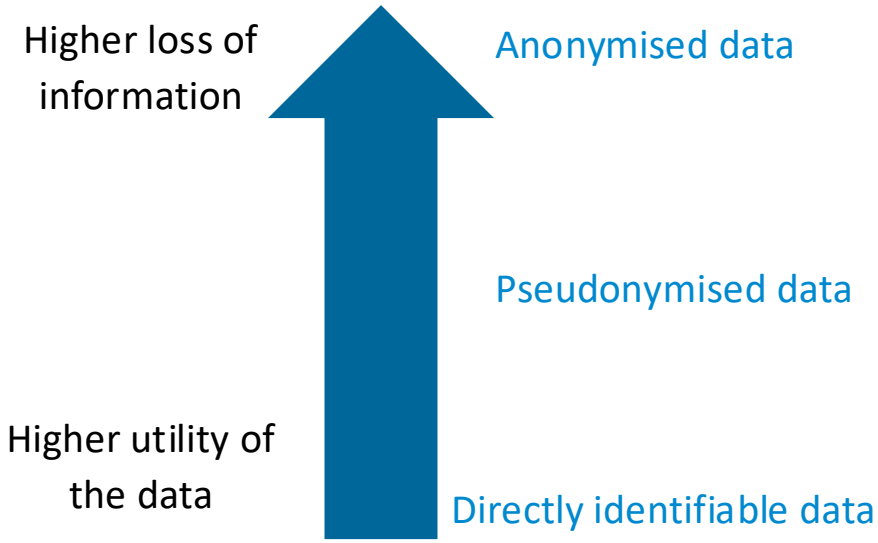
They are as effective as the other measures that are taken alongside it

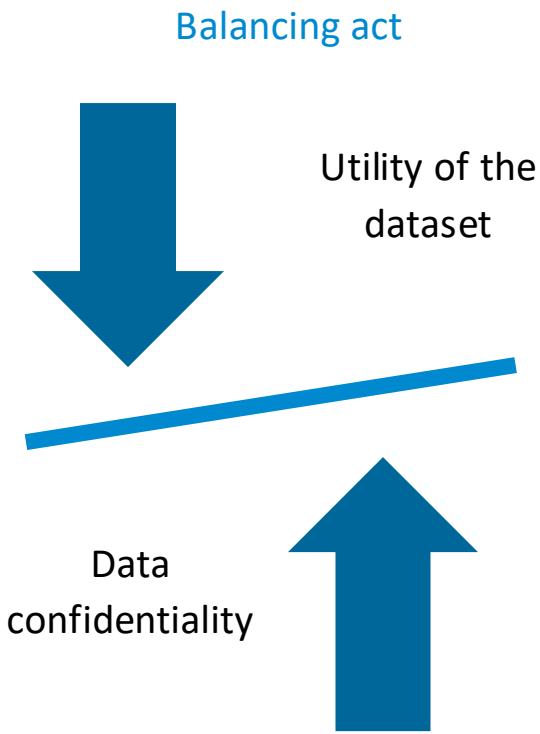
Access control

Data minimization

File encryption

De-identification is a balancing act





Key concepts: Data minimisation

Data minimisation

The best way to deidentify data, is to think ahead of time whether you really need certain data categories to be collected

De-identification in practice

Who’s who?

Sam
Alex
Peter
Maria
Anne
Claire
Philip
Alfred
Bill
Herman
Anita
Paul
Frans
Max
Richard
Bernard
Robert
Susan
Charles
David
Tom
Joe
George
Eric

Final notes

1.Context Matters: The re-identification risk depends heavily on the specific data, the available background knowledge, and the attacker's capabilities.

2. No Absolute Guarantee: No de-identification method is foolproof. There's always a risk of re-identification, especially with increasingly sophisticated data analysis techniques.

3. False promises: Do not promise that data will be fully anonymous if you do not have proof to show the participant that it will be truly anonymous or if you don't have an anonymization strategy to show.

4. Combining Techniques: Using multiple techniques in combination provides stronger protection. But do not go to far as to make your data unusable for your research goal.

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Join us at the next FGB Data Coffee Break!

- **Topic:** Do's and Don't's of DMP's
- **Time** : 15hs – 15.45hs
- **Location:** MF-D134- Salutem 2

- **Registration:**

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Thank you for joining!

- Monthly meetings about Research Data
- Management by the Research Policy and Support (RePS) team of FGB
- Short, focused and informal
- Monthly sessions
- Do you have suggestions for a topic? **research.data.fgb@vu.nl**

Acknowledgements

- Annika Valaranta. (2021). How to guide anonymisation?. Global Virtual Conference (IASSIST 2021), Online. Zenodo. https://doi.org/10.5281/zenodo.5181069
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